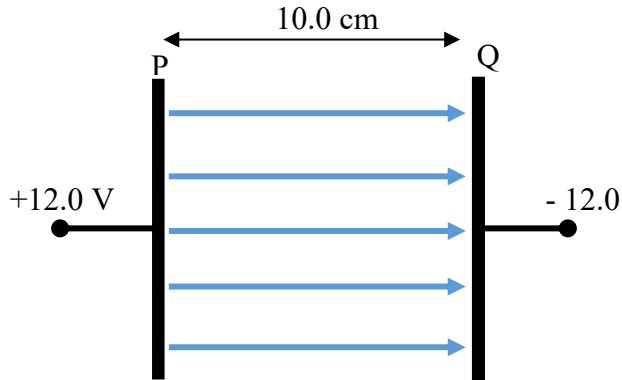


Q5(a)



Q5(b)(i)

$$E = \frac{\Delta V}{d} = \frac{12 - (-12)}{10.0 \times 10^{-2}} \\ = 240 \text{ V m}^{-1}$$

$$\text{Net Force} = Eq = ma$$

$$a = \frac{Eq}{m} = \frac{240 \times 1.6 \times 10^{-19}}{9.11 \times 10^{-31}} = 4.22 \times 10^{13} \text{ m s}^{-2}$$

Time to cover 0.15 m of plate is

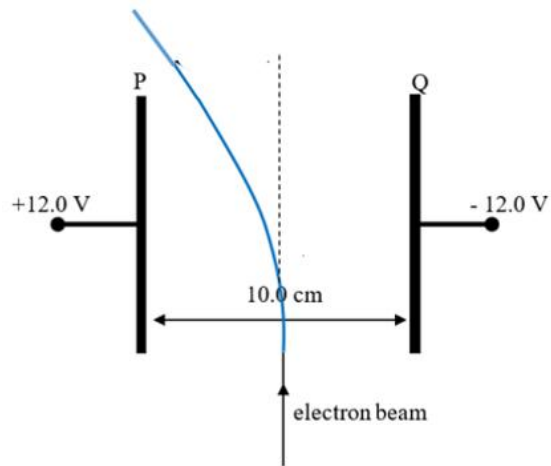
$$t = \frac{0.15}{4.5 \times 10^6} = 3.33 \times 10^{-8} \text{ s}$$

Using $s = ut + \frac{1}{2}at^2$ for the horizontal motion of the electron, we have

$$s_x = \frac{1}{2}(4.22 \times 10^{13})(3.33 \times 10^{-8})^2 \\ = 0.0234 \text{ m}$$

Since this is smaller than 5.0 cm from midway between plates to a plate, the electrons will clear the plates.

Q5(b)(ii)



Assumption: Gravitational and magnetic field effects due to the Earth are negligible.

Q5(c)

The proton beam will be able to clear the plates.

The mass of the proton is much larger than an electron while its charge is the same magnitude as that of the electron. The proton will experience a smaller horizontal electric acceleration and will not deflect as much as an electron with the same velocity.

Q6(c)(i)